INCH-POUND
MIL-DTL-24643/11F
27 November 2023
SUPERSEDING
MIL-DTL-24643/11E
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LSSSF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First - Copper conductor, uncoated, 300 MCM, ASTM B173, class G stranding (see note 2).

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Thermoset insulation. Minimum average wall thickness of 0.075 inch. Insulation color shall be black or natural.

Fourth - Cross-linked polyolefin jacket bonded to the underlying insulation. Absolute minimum average wall thickness of 0.075 inch. See table I for cable overall diameter.

NOTE 1: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643). The identification tape information must be accessible and legible.

NOTE 2: The copper shall be hard drawn in accordance with ASTM B173, having a minimum tensile strength of 67,200 lb/in and minimum elongation of 0.85 percent in 60-inch lengths before stranding.

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/11	Type and size	Overall diameter (inches)	
		min.	max.
-01UN	LSSSF-300	1.020	1.100

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	0.0388
Voltage withstand (volts, root mean square, min.)	
Conductor to ground	2,500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to ground	100
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Cold bending (cable) at -22±3.6 °F (-30±2 °C) (two turns around an 8.5-inch maximum mandrel diameter)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

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Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Insulation	45
Toxicity index (max.)	
Jacket	5
Insulation	1.5

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Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (min.)	Feet (max.)
LSSSF-300	1,045	1,155

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-027)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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